

MD. SHAHARIAR HASSAN

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RESEARCH & ACADEMIC INTERESTS

High-Speed Networking & Embedded Systems: DPDK-based packet processing, Linux kernel bypass, low-latency systems, FPGA-to-host communication, and embedded software.

Digital Systems & Computer Architecture: RISC-V microarchitecture, pipelined processors, FPGA prototyping, digital IC design, and hardware verification.

Wireless Systems & Security: Physical-layer security, wireless channel modeling, high-throughput telecom systems, and next-generation communication infrastructure.

EDUCATION

Rajshahi University of Engineering & Technology (RUET), Bangladesh | B.Sc. in Electrical & Computer Engineering (ECE), Jan 2018 - Oct 2023

CGPA: 3.21 / 4.00 | **Dual Track:** Electrical & Electronic Engineering (EEE) and Computer Science & Engineering (CSE)

Undergraduate Thesis: Analyzing Secrecy Performance in Wireless Communication Networks with Randomly Distributed Eavesdroppers

Supervisor: Milton Kumar Kundu, Assistant Professor, Department of ECE, RUET. Studied secrecy outage probability and achievable secrecy rates using stochastic geometry, Poisson point process models, and wireless fading channels.

Undergraduate Review Paper: A Review on Wearable Technologies in Early Detection of Lymphedema. Reviewed non-invasive wearable sensing technologies and data-driven approaches for continuous monitoring and early detection.

Selected Coursework: EEE - Circuits & Systems, Analog Electronics, Electrical Machines, Power Systems, Communication Engineering, Digital Signal Processing, Control Systems, VLSI Design. CSE - C/C++ Programming, OOP, Data Structures & Algorithms, Database Systems, Operating Systems, Computer Architecture, Software Engineering, Microprocessors & Assembly Language.

STANDARDIZED TEST

IELTS Academic: Overall 6.5 | Listening 6.0 | Reading 6.0 | Writing 6.5 | Speaking 7.0 | September 2026

TECHNICAL SKILLS

High-Speed Networking & Linux: DPDK, Kernel Bypass, HugePages, CPU Pinning/Isolation, POSIX Shared Memory, TAP Interfaces, dpdk-testpmd, Zero-Copy Processing

Digital Design & Architecture: Verilog HDL, SystemVerilog, RISC-V, Pipelined Datapaths, Hazard Detection, Forwarding, FIFO, FSM, ALU

Verification & Automation: Cocotb, Verilator, Icarus Verilog, GTKWave, Python Testbenches, Makefile Automation, GitHub Actions, GitLab CI, Docker

FPGA & EDA: AMD Xilinx Vivado, RTL Simulation, Synthesis, Implementation, Timing Analysis, XDC, Artix-7 FPGA

Embedded Systems: Bare-Metal C/C++, STM32 ARM Cortex-M, AVR/Arduino, ESP32, UART, SPI, I2C, gRPC

PCB & Laboratory: KiCad, Multisim, Proteus, Digital Oscilloscope, Logic Analyzer, PCB Design & Debugging

PROFESSIONAL ENGINEERING EXPERIENCE

Siliconova Ltd., Dhaka, Bangladesh | *Embedded Software Engineer I* | July 2024 - Present

- Develop high-speed C-based data-plane software using DPDK, HugePages, CPU isolation, and zero-copy packet processing.
- Design, test, and integrate Verilog RTL modules on Xilinx Artix-7 FPGAs, including high-speed Ethernet communication between FPGA hardware and Linux host systems.
- Use AMD Xilinx Vivado, Docker, CI/CD, and Intel VTune Profiler for implementation, automation, system analysis, and performance optimization.
- Achieved sustained 100 Gbps packet throughput with zero packet loss during end-to-end system validation.
- Develop gRPC services in Python and C for hardware configuration, register access, link control, and system monitoring.
- Collaborate with international engineering teams and contribute to high-throughput telecom projects for XRComm, USA, covering debugging, validation, optimization, and project delivery.

Selected Telecom Systems Delivered: 4T4R MIMO (1.966 GHz aggregate); 3T3R MIMO (1.476 GHz); Wideband 1-CH (1.4756 GHz); Wideband 2-CH (2.951 GHz); Wideband 4-CH (5.902 GHz).

Additional architecture work: Performance planning for higher-bandwidth FPGA-to-host systems using 400G NIC, PCIe, and QSFP28 interfaces.

SELECTED HARDWARE & RESEARCH PROJECTS

3-Stage Pipelined RISC-V Core & Automated Verification | *Verilog* | *Cocotb* | *Verilator* | *GTKWave*

- Refactored a single-cycle PicoRV32 design into a 3-stage pipelined RISC-V processor with IF, ID/EX, and MEM/WB stages.
- Implemented hazard detection, data forwarding, load-use stall handling, and branch flushing.
- Developed cycle-accurate Cocotb testbenches and automated regression testing using Makefiles.

Digital Logic Blocks & Verification Suite | *Verilog* | *Cocotb* | *Icarus Verilog*

- Designed and verified configurable synchronous FIFO buffers with full, empty, almost-full, and almost-empty indicators.
- Developed Moore and Mealy FSM controllers and verified read/write, overflow, underflow, and data-integrity behavior.

Microcontroller System Board & Multi-Sensor Interface | *KiCad* | *Multisim* | *Bare-Metal C*

- Designed schematics and a 2-layer PCB for a microcontroller-based multi-sensor system.
- Developed interrupt-driven UART and SPI drivers and tested power, clock, and signal integrity using laboratory equipment.

ACADEMIC REFEREES

Milton Kumar Kundu - Assistant Professor, Department of ECE, RUET | Thesis Supervisor |
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Md. Robiul Islam - Assistant Professor, Department of ECE, RUET | Academic Referee |
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